Verification & Validation

1 Fault Diagnosis

Audio Analysis Using Our Drone Audio	Blackbox Analysis Using Our Drone Flight Data
Accuracy: 0.815/1	Accuracy: 0.868/1
F1 Score: 0.715/1	F1 Score: 0.929/1

$$\text{Precision (P)} = \frac{\text{True Predicated Positives}}{\text{All Predicated Positives}} \qquad \text{Recall (R)} = \frac{\text{True Predicated Positives}}{\text{All Real Positives}}$$

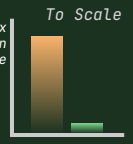
F1 Score: Harmonic Mean of Precision and Recall

2 Diagnostic Reliability

Verified under **noisy environments** to simulate varying conditions, and maintained **consistent** results.

3 Operational Integration

(Analysis Speed)



Betaflight website serial log extraction: **~5 min**
Mass storage mode extraction from device: **<30 sec**

4 Field Portability

Size of **handheld** multimeter, runs off a **power bank**, stores data **locally**

5 Diagnostic Traceability

Every test was checked for **logging completeness**; MicroSD with **customizable** storage; Data **synced** to cloud database

Validated workflow through real-world **beta testing**: Clover can be operated with minimal training.

Sustainability: Individual components are **easily replaceable**. Code can be **open-sourced** for customization

Future Work

- **More Testing:** Test across D. Vision Aerials' full drone fleet, in windy and cold conditions.
- **Improved Analysis:** Expand blackbox analytics (battery monitoring, PID tuning) and improve algorithm speed.
- **Enterprise System:** Add user accounts for fleet-wide diagnostic tracking.
- **Additional Sensors:** Extend diagnostics with more sensors (e.g., thermal monitoring).

Team 15

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Clover

Drone Diagnostics

Filming at a concert
Credit: D. Vision Aerials

Problem Context

- Drones sometimes operated in **densely populated areas**
- Even minor **motor faults** can cause crashes and damage
- Current fault detection methods relies on **subjective** operator sound checks

Opportunity

A portable system to verify the **health and flight readiness** of drone **propulsion** systems using **objective** pre-flight diagnostics.

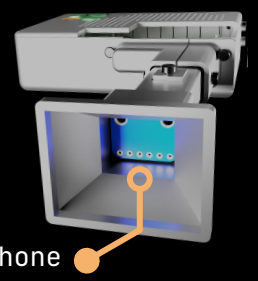
Objectives

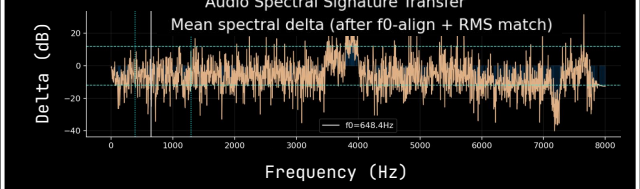
1. Fault Diagnosis
2. Diagnostic Reliability
3. Operational Integration
4. Field Portability
5. Diagnostic Traceability

Clover is a **portable drone diagnostic device** that analyzes motor acoustics and flight telemetry to assess **propulsion system** (motors, propellers…) **health** before flight.

Step 1: Audio Analysis

Start Session → Record Audio → Analyze Audio





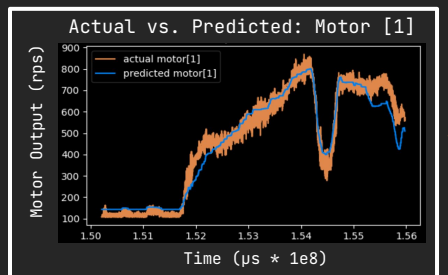
Graph 1: Sample data output from microphone after audio transformation to be analysed by neural network.

- ▶ Clover microphone records drone audio.
- ▶ Differences between the **recorded audio** and the **baseline audio** are used to produce the transformed audio.
- ▶ The transformed audio is processed by a deep neural network trained on 324k audio samples from Yi et al (2023)'s* drone audio dataset.

* W. Yi, J.-W. Choi, and J.-W. Lee, "Sound-based drone fault classification using multitask learning," arXiv preprint arXiv:2304.11708, 2023.

Step 2: Blackbox

Record Blackbox → Analyze Blackbox

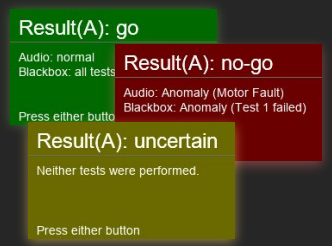


Graph 2: Shows exported blackbox data from 1/4 motors and how it maps onto the data from a healthy drone

- ▶ Clover connects to the drone blackbox and receives past flight data.
- ▶ Differences between expected and actual motor output indicate a fault.

Step 3: Results

Display Results → Erase Flash → End Session



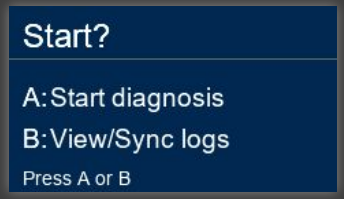
- ▶ Audio and blackbox analysis are combined to produce a final go/no-go/caution result displayed on screen.

- ▶ [Optional] Clover clears onboard flash memory to prevent log storage conflicts.

- ▶ Clover generates a 6-digit Pair Code to securely link the device with the web/mobile app.

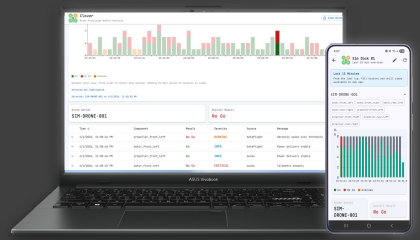
Step 4: Logging

Sync Logs → View Logs



supabase

- ▶ Logs are first stored in the local Raspberry Pi microSD card (64GB) and then synced to the cloud database.



- ▶ Diagnostic dashboards on the Clover web/mobile app displays the final result, severity level and historical trends.